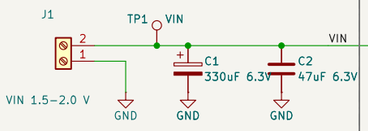
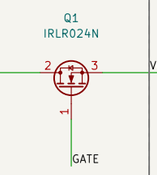
Input

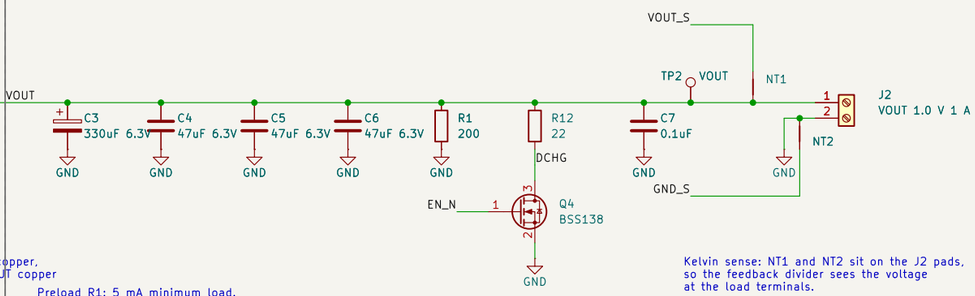


Pass device



Drain tab on the VIN copper,
source lead to the VOUT copper

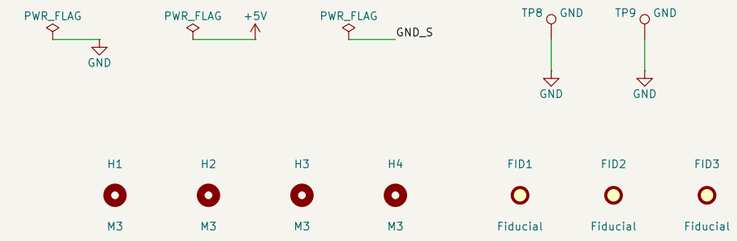
Output, preload, discharge and Kelvin sense



Preload R1: 5 mA minimum load.
Q4 and R12 discharge the output when EN is low.

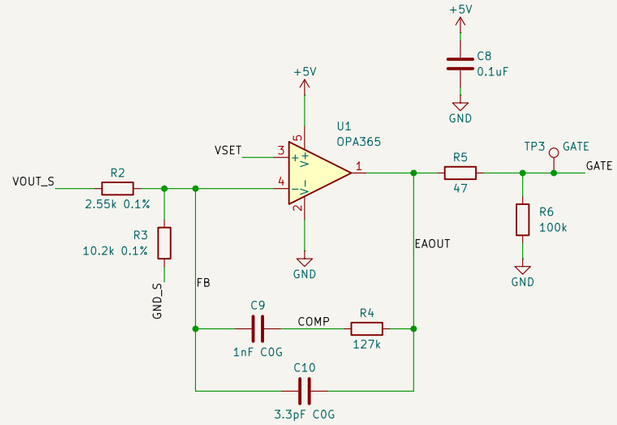
Kelvin sense: NT1 and NT2 sit on the J2 pads,
so the feedback divider sees the voltage
at the load terminals.

Power flags, test points and mechanical



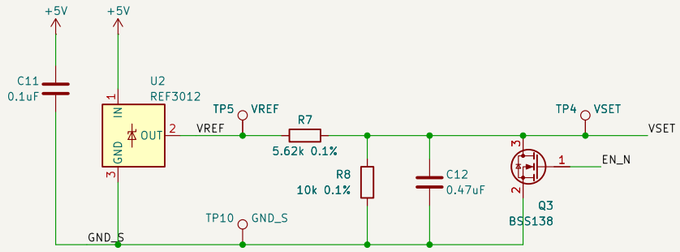
Test points: Keystone 5005 red (VIN, VOUT, +5V), 5007 white (GATE, VSET, VREF, EN_INT),
5006 black (GND, GND_S).

Error amplifier and compensation



PI compensation: Kp = R4/R2 = 50, zero 1/(2 pi R4 C9) = 1.25 kHz,
pole 1/(2 pi R4 C10) = 380 kHz. R5 isolates U1 from about 2 nF of gate
capacitance in the model. D1 and R13 (enable block) pull FB up when the regulator is off.

Reference and soft start



VSET = 1.25 V x R8/(R7+R8) = 0.800 V
VOUT = VSET x (1 + R2/R3) = 1.000 V
Soft start and reference filter: tau = (R7 || R8) x C12 = 1.69 ms
Q3 empties C12 while EN_N is high, so every start is a soft start.
U2, C11, R3, R8, C12 and Q3 return to GND_S, the quiet ground
tied to J2 by NT2.

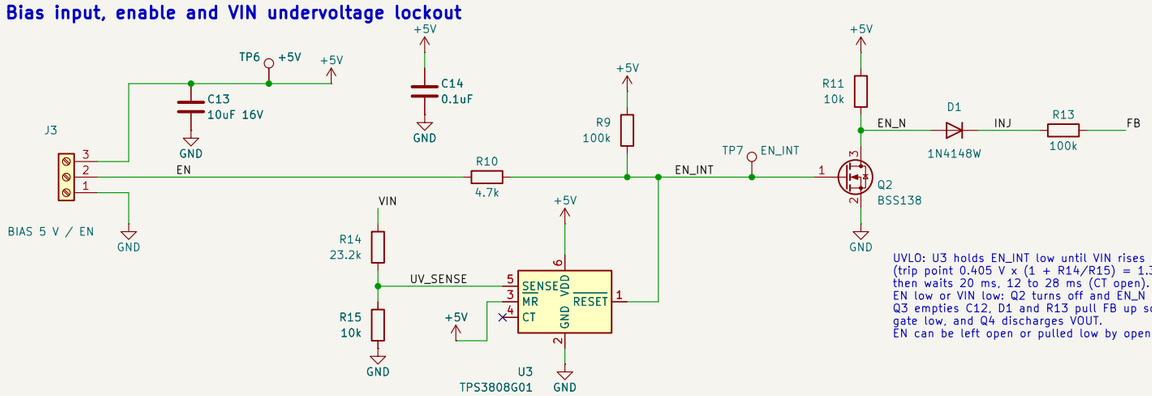
Design summary

Input 1.5 V to 2.0 V (nominal 1.8 V), output 1.000 V, 0 to 1 A.
The 5 V bias powers U1, U2 and U3 and gives the NMOS follower
its gate drive headroom: about 3.2 V on the gate at 1 A and up
to 4.98 V near dropout.
Q1 dissipates up to 1.0 W at VIN 2.0 V and 1 A.
Output capacitors: one 330 uF polymer and three 47 uF X5R.
Simulated results and the design math are in README.md.

Connectors

J1 pin 1 GND, pin 2 VIN (1.5 V to 2.0 V)
J2 pin 1 VOUT (1.000 V, 0 to 1 A), pin 2 GND
J3 pin 1 GND, pin 2 EN (open or high = on), pin 3 +5V bias
All three are Phoenix Contact MKDS 1,5 screw terminals, 5.08 mm pitch.

Bias input, enable and VIN undervoltage lockout



UVLO: U3 holds EN_INT low until VIN rises past about 1.37 V
(trip point 0.405 V x (1 + R14/R15) = 1.345 V plus 1.5 % hysteresis),
then waits 20 ms, 12 to 28 ms (CT open). It trips again below 1.345 V.
EN low or VIN low: Q2 turns off and EN_N goes high.
Q3 empties C12, D1 and R13 pull FB up so U1 drives the
gate low, and Q4 discharges VOUT.
EN can be left open or pulled low by open drain logic.

VIN 1.5–2.0 V, VOUT 1.000 V, 0–1 A, 5 V bias, Kelvin sense at J2
Discrete NMOS low dropout regulator for a low voltage core rail

Siyun Chen

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Title: NMOS LDO board, 1.0 V at 1 A

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1